

ECE 313: Probability with Engineering Applications

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Homework 12

Name: _____

Student ID: _____

Due Date: January 2 23:59, 2026

Problem 1. Suppose X, Y have joint pmf

$$f_{X,Y}(x,y) = \begin{cases} c(x+y), & 0 \leq x \leq 2, 0 \leq y \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the normalizing constant c .
- (b) Find the unconstrained MMSE estimate of X given $Y = 1$, i.e. $\mathbb{E}[X | Y = 1]$.
- (c) Find the best linear estimator of X given $Y = y_0$.

Problem 2. Suppose two random variables X and Y have the following joint pdf:

$$f_{X,Y}(u,v) = \begin{cases} \frac{uv+1}{4} & \text{if } -1 \leq u \leq 1 \text{ and } -1 \leq v \leq 1, \\ 0 & \text{else} \end{cases}$$

- (a) Find the numerical value of $P\{Y \leq -\frac{1}{3}\}$.
- (b) Find the constant estimator δ^* of Y and the corresponding mean square error (MSE).
- (c) Find the unconstrained estimator $g^*(X)$ for observation X and the corresponding MSE.
(Hint: The MSE must be deterministic, not random.)
- (d) Find the linear estimator $L^*(X)$ for observation X and the corresponding MSE.

Problem 3. Given the following likelihood matrix, specify the ML and MAP decision rules and find $p_{\text{false-alarm}}$, p_{miss} , and p_e for both of them. Assume that:

$$\pi_0 = 0.3.$$

	$X = 0$	$X = 1$	$X = 2$	$X = 3$	$X = 4$
H_1	0.00	0.10	0.24	0.34	0.32
H_0	0.08	0.15	0.31	0.31	0.15

Problem 4. On the basis of a sensor output X , it is to be decided which hypothesis is true:

$$H_0 \text{ vs. } H_1.$$

If H_0 is true then X has density f_0 , and if H_1 is true then X has density f_1 , where

$$f_0(u) = \begin{cases} \frac{1}{2}, & |u| \leq 1, \\ 0, & |u| > 1, \end{cases} \quad f_1(u) = \begin{cases} |u|, & |u| \leq 1, \\ 0, & |u| > 1. \end{cases}$$

- (a) Describe the maximum-likelihood (ML) decision rule.
- (b) Compute the false-alarm probability $P_{\text{false alarm}}$ and the miss probability P_{miss} for the ML rule.
- (c) Assume a priori that H_0 is true with probability π_0 and H_1 is true with probability $\pi_1 = 1 - \pi_0$. For what values of π_0 does the MAP rule declare H_1 with probability one, no matter which hypothesis is actually true?
- (d) Under the same priors, for what values of π_0 does the MAP rule declare H_0 with probability one, no matter which hypothesis is actually true?